Q1 2021 (31 Aug Shift 2)

If velocity $[V]$, time $[T]$ and force $[F]$ are chosen as the base quantities, the dimensions of the mass will be :

(1) $\left[FT^{-1} V^{-1}\right]$

(2) $\left[FTV^{-1}\right]$

(3) $\left[FT^{2} V\right]$

(4) $\left[FVT^{-1}\right]$

Q2 2021 (31 Aug Shift 1)

Which of the following equations is dimensionally incorrect?

Where $t =$ time, $h =$ height, $s =$ surface tension, $\theta =$ angle, $\rho =$ density, a, $r =$ radius, $g =$ acceleration due to gravity, $v =$ volume, $p =$ pressure, $W =$ work done, $\Gamma =$ torque, $\in=$ permittivity, $E =$ electric field, $J =$ current density, $L =$ length.

(1) $v = \frac{\pi p a^{4}}{8 \eta L}$

(2) $h = \frac{2 s \cos \theta}{\rho r g}$

(3) $\mathbf{J} = \in \frac{\partial E}{\partial t}$

(4) $W = \Gamma \theta$

Q3 2021 (31 Aug Shift 1)

Match List-I with List-II.

List-I List-II

(a) Torque

(i) MLT^{-1}

(b) Impulse

(ii) MT^{-2}

(c) Tension

(iii) $ML^{2} T^{-2}$

(d) Surface Tension

(iv) MLT^{-2}

Choose the most appropriate answer from the option given below :

(1) (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)

(2) (a)-(ii), (b) –(i), (c)-(iv), (d)-(iii)

(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(4) (a)-(iii), (b)-(iv), (c) -(i), (d)-(ii)

Q4 2021 (27 Aug Shift 2)

If force (F), length (I.) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density :

(1) $\left[FL^{-4}T^{2}\right]$

(2) $\left[FL^{-3}T^{2}\right]$

(3) $\left[FL^{-5}T^{2}\right]$

(4) $\left[FL^{-3}T^{3}\right]$

Q5 2021 (27 Aug Shift 2)

Match List-I with List-II.

List-I List-II

(a) R_H (Rydberg constant)

(i) $kgm^{-1}s^{-1}$

(b) h (Planck's constant)

(ii) $kgm^{2}s^{-1}$

(c) μ_B (Magnetic field

(iii) m^{-1}

energy density)

(d) η (coefficient of viscocity)

(iv) $kgm^{-1} s^{-2}$

Choose the most appropriate answer from the options given below :

(1) (a) −(ii), (b) − (iii), (c)-(iv), (d)-(i)

(2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(3) (a) −(iv), (b) − (ii), (c) − (i), (d)− (iii)

(4) (a) −(iii), (b) − (ii), (c)-(i), (d)-(iv)

Q6 2021 (27 Aug Shift 1)

If E and H represents the intensity of electric field and magnetising field respectively, then the unit of E/H will be :

(1) ohm

(2) mho

(3) joule

(4) newton

Q7 2021 (27 Aug Shift 1)

Which of the following is not a dimensionless quantity?

(1) Relative magnetic permeability (μ_r)

(2) Power factor

(3) Permeability of free space (μ_0)

(4) Quality factor

Q8 2021 (26 Aug Shift 2)

Match List-I with List-II.

\multicolumn2|c|List − I	\multicolumn2c|List − II		
(a)	Magnetic Induction	(i)	$ML^2T^{-2}A^{-1}$
(b)	Magnetic Flux	(ii)	$M^0L^{-1}A$
(c)	Magnetic Permeability	(iii)	$MT^{-2}A^{-1}$
(d)	Magnetization	(iv)	$MLT^{-2}A^{-2}$

Choose the most appropriate answer from the options given below:

(1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(2) (a)-(ii) , (b) − (i), (c) − (iv), (d) − (iii)

(3) (a) −(iii), (b) − (ii), (c) − (iv), (d) − (i)

(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q9 2021 (26 Aug Shift 1)

If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimensions of P in the formula $P = EL^2M^{-5}G^{-2}$ are :-

(1) $[M^0L^1T^0]$

(2) $[M^{-1}L^{-1}T^2]$

(3) $[M^1L^1T^{-2}]$

(4) $[M^0L^0T^0]$

Answer Key

Q1 (2)	**Q2** (1)	**Q3** (1)	**Q4** (1)
Q5 (2)	**Q6** (1)	**Q7** (3)	**Q8** (4)
Q9 (4)			

Q1 (2)

$[\mathrm{M}]=\mathrm{K}[\mathrm{F}]^{\mathrm{a}}[\mathrm{T}]^{\mathrm{b}}[\mathrm{V}]^{\mathrm{c}}$

$\left[\mathrm{M}^{1}\right]=\left[\mathrm{M}^{1}\ \mathrm{L}^{1}\ \mathrm{T}^{-2}\right]^{\mathrm{a}}\left[\mathrm{T}^{1}\right]^{\mathrm{b}}\left[\mathrm{L}^{1}\ \mathrm{T}^{-1}\right]^{\mathrm{c}}$

$\mathrm{a}=1,\ \mathrm{b}=1, \mathrm{c}=-1$

$\therefore[\mathrm{M}]=\left[\mathrm{FTV}^{-1}\right]$

Q2 (1)

(i) $\frac{\pi p a^{4}}{8 \eta L}=\frac{d v}{d t}=$ Volumetric flow rate (poiseuille's law)

(ii) $\mathrm{h} \rho \mathrm{g}=\frac{2\,\mathrm{s}}{\mathrm{r}} \cos \theta$

(iii) RHS $\Rightarrow \varepsilon \times \frac{1}{4 \pi \varepsilon_{0}} \frac{\mathrm{a}}{\mathrm{r}^{2}} \times \frac{1}{\varepsilon}=\frac{\mathrm{q}}{\mathrm{t}} \times \frac{1}{\mathrm{r}^{2}}=\frac{\mathrm{I}}{\mathrm{L}^{2}}=\mathrm{IL}^{-2}$

LHS

$$\mathrm{T}=\frac{\mathrm{I}}{\mathrm{A}}=\mathrm{IL}^{-2}$$

(iv) $\mathrm{W}=\tau \theta$

Option (1)

Q3 (1)

torque $\tau \rightarrow \mathrm{ML}^{2}\ \mathrm{T}^{-2}$ (III)

Impulse $\mathrm{I} \Rightarrow \mathrm{MLT}^{-1}$ (I)

Tension force $\Rightarrow \mathrm{MLT}^{-2}$ (IV)

Surface tension $\Rightarrow \mathrm{MT}^{-2}$ (II)

Option (1)

Q4 (1)

$$\text{Density} = \left[F^a L^b T^c\right]$$
$$\left[ML^{-3}\right] = \left[M^a L^a T^{-2a} L^b T^c\right]$$
$$\left[M^1 L^{-3}\right] = \left[M^a L^{a+b} T^{-2a+c}\right]$$
$$a = 1; \quad a + b = -3; \quad -2a + c = 0$$
$$1 + b = -3 \quad c = 2a$$
$$b = -4 \quad c = 2$$

So, density $= \left[F^1 L^{-4} T^2\right]$

Q5 (2)

SI unit of Rydberg const. $= m^{-1}$

SI unit of Plank's const. $= kgm^2\ s^{-1}$

SI unit of Magnetic field energy density $= kgm^{-1}\ s^{-2}$

SI unit of coeff. of viscosity $= kgm^{-1}\ s^{-1}$

Q6 (1)

Unit of $\frac{E}{H}$ is $\frac{\text{volt / metre}}{\text{Ampere / metre}} = \frac{\text{volt}}{\text{Ampere}} = \text{ohm}$

Q7 (3)

$$[\mu_r] = 1 \text{ as } \mu_r = \frac{\mu}{\mu_m}$$

[power factor $(\cos\phi)] = 1$

$\mu_0 = \frac{B_0}{H}\left(\text{ unit } = NA^{-2}\right)$: Not dimensionless

$$[\mu_0] = \left[MLT^{-2}\ A^{-2}\right]$$

quality factor $(Q) = \frac{\text{Energy stored}}{\text{Energy dissipated per cycle}}$

So Q is unitless \& dimensionless.

Q8 (4)

(a) Magnetic Induction $= \mathrm{MT}^{-A}\mathrm{A}^{-}$

(b) Magnetic Flux $= \mathrm{ML}^{2}\,\mathrm{T}^{-2}\,\mathrm{A}^{-1}$

(c) Magnetic Permeability $= \mathrm{MLT}^{-2}\,\mathrm{A}^{-2}$

(d) Magnetization $= \mathrm{M}^{(0}\mathrm{L}^{-1}\,\mathrm{A}$

Ans. (4)

Q9 (4)

$$\mathrm{E} = \mathrm{ML}^{2}\,\mathrm{T}^{-2}$$
$$\mathrm{L} = \mathrm{ML}^{2}\,\mathrm{T}^{-1}$$
$$\mathrm{m} = \mathrm{M}$$
$$\mathrm{G} = \mathrm{M}^{-1}\,\mathrm{L}^{+3}\,\mathrm{T}^{-2}$$
$$\mathrm{P} = \frac{\mathrm{EL}^{2}}{\mathrm{M}^{5}\mathrm{G}^{2}}$$
$$[\mathrm{P}] = \frac{\left(\mathrm{ML}^{2}\,\mathrm{T}^{-2}\right)\left(\mathrm{M}^{2}\,\mathrm{L}^{4}\,\mathrm{T}^{-2}\right)}{\mathrm{M}^{5}\left(\mathrm{M}^{-2}\,\mathrm{L}^{6}\,\mathrm{T}^{-4}\right)} = \mathrm{M}^{0}\,\mathrm{L}^{0}\,\mathrm{T}^{0}$$

Option (4)